Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Teacher's Edition

The Period

MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

Must Decide whether three lengths make a triangle (the two shorter sum to more than the longest) and a right triangle (the converse, on the longest side).

Must not Acute or obtuse by comparing squares (Geometry); sides as expressions; finding a missing side (6.03).

Target I can decide whether three lengths can make a triangle, and whether they make a right triangle.

MTR **MTR.2.1** — Notes I: three lengths as sticks and as one inequality · **MTR.4.1** — boards 3, 5 and 7 split; the sixty-second re-vote · **MTR.6.1** — board 7: a reason checked against the squares, not the sum

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Add the two shorter sides: MORE than the longest is a triangle; equal lies flat.
2. The third side is between the difference and the sum.
3. Right triangle: the shorter sides' squares add up to the longest side's square.

Slide by Slide

1. **Triangle Inequality Theorem; Converse of the Pythagorean Theorem.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q3 and Q4 are a pair: $6^2 + 8^2 = 100 = 10^2$, today's converse before it has a name. *1 retrieves 5.10–11 — a unit rate; yesterday was the test, so the band reaches the last lesson (yesterday). 2 retrieves 5.02–03 — area scales by k^2 (last week). 3 retrieves 4.05 — a perfect square's root (Unit 4) (two units back). 4 retrieves MA.7.NSO.1.1 — whole-number powers; the converse adds two squares (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 288 mi 2. 45 cm^2 3. 10 4. 100
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule.** Equal is not enough. Equal lies flat.
6. **Notes.** Two sides, 4 and 9.
7. **Notes.** Circle the longest side first. It is the only one that can be the hypotenuse.
8. **Example 1 — Can the three lengths make a triangle?** Ask which two sides get added before anyone adds: the two shorter, 5 and 8.
9. **Example 1 — Worked.** 5 + 8 = 13, and 13 is less than 14: the two short sides cannot reach across.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 7 + 9 = 16 > 15, so yes — a board that says no saw the big 15 and did not add.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Is it a right triangle?** Ask which side is c before anyone squares: 20, the longest, though it is listed first.
13. **Example 2 — Worked.** The longest side is 20, listed first.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 64 + 121 = 185, not 196, so not right — a board that says 'right' added the sides, 8 + 11 = 19 > 14: the Triangle Inequality, not the converse (board 9).
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. Can sides of 2 ft, 7 ft and 10 ft make a triangle? → ***No — 2 + 7 = 9 is less than 10.*** *Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]; 'yes — the ends just do not meet' [Florida B1G-M, GR.1.3: believing the sides need not meet at the corners]*

18–19. Board 2. Can sides of 6 m, 6 m and 11 m make a triangle? → ***Yes — 6 + 6 = 12 is more than 11.*** *No — 'two sides cannot be equal' [Notes I: only the sum matters]; 'it is not a right triangle, so the rule does not apply' [Florida B1G-M, GR.1.3: the Triangle Inequality holds for every triangle]*

20–21. Board 3. Which three lengths can make a triangle? → ***B — 5, 11, 8*** *(A) 7 + 7 = 14 counted — equal lies flat [Notes I]; (C) added the first two listed, 3 + 10 [MA.8.GR.1.3: test the two shorter]; (D) 4 + 6 = 10 is less than 12: those sides cannot meet [Florida B1G-M, GR.1.3: believing the sides need not meet at the vertices] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

22–23. Board 4. Two sides of a triangle are 7 cm and 12 cm. Write an inequality for the length x, in centimeters, of the third side. → ***5 < x < 19 — more than 5 cm***

and less than 19 cm. *5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]; x < 19 — no bottom end [Notes II: long enough to reach]*

24–25. Board 5. Two sides of a triangle are 6 cm and 10 cm. Which inequality shows every possible length x, in centimeters, of the third side? → ***C — 4 < x < 16*** *(A) between the two given sides — the third side can be shorter than both [Notes II: more than the difference]; (B) the flat cases included: at 4 or 16 the sides lie flat [Notes II: never equal]; (D) no bottom end — a third side of 4 or less cannot reach [Notes II: more than the difference] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. Do sides of 8 inches, 15 inches and 17 inches make a right triangle? → ***Yes — $8^2 + 15^2 = 289 = 17^2$.*** *No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]*

28–29. Board 7. A triangle has sides of 25 cm, 7 cm and 24 cm. Is it a right triangle? Which answer has the right reason? → ***A — yes, because $7^2 + 24^2 = 25^2$*** *(B) the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]; (C) 25 taken as a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) added the sides, not their squares [Notes III] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

30–31. Board 8. A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds. Does the sail have a right angle? → ***Yes — $20^2 + 21^2 = 841 = 29^2$.*** *The 12 pounds is not needed. 'Cannot tell without the angles' [Notes III: the sides are enough]*

32–33. Board 9. Ana says sides of 6, 7 and 9 make a right triangle because 6 + 7 > 9. Is she right? Explain. → ***No — 6 + 7 > 9 shows only a triangle; 85 is not 81, so it is not right.*** *'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: a triangle? (no, then yes with two equal sides); 3 votes with the equal case and the wrong pair; 4–5 the third side's range, written then voted. The pattern breaks at 6 — a RIGHT triangle?, squares not sums — and 7 lists the longest side first; 8 has the extra number; 9 is written. The independent set walks the same path.

Misconceptions to Watch

Board 1. Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]

Board 2. No — 'two sides cannot be equal' [Notes I: only the sum matters]

Board 3. 7 + 7 = 14 counted — equal lies flat [Notes I]

Board 4. 5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]

Board 5. between the two given sides — the third side can be shorter than both [Notes II: more than the difference]

Board 6. No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]

Board 7. the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]

Board 8. 'Cannot tell without the angles' [Notes III: the sides are enough]

Board 9. 'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]

Materials Whiteboards and markers. Optional: strips cut to 3, 4, 5, 6 and 9 units for Notes I. Calculators allowed, printed nowhere.